Development of a Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management

by

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CHAPTER 1

INTRODUCTION

1.1. Background of the Study

Local Government Units (LGUs) in the Philippines have been on the front line in promoting digital transformation in public service delivery in recent years [1]. On a more intimate level, the barangay is the smallest unit implementing the expected government functions. Most barangays still process documents, organize files, and manage resident information in manual and paper-based systems. These conventional methods, though familiar, are often inefficient and time-consuming, especially during peak seasons when the demand for barangay documents significantly increases

Digitalization of service delivery, however, has its disadvantages. Services are usually slow, apathetic, and painfully long, especially during the peak seasons. Residents sometimes have to wait for hours to process a simple document request.[2]. The current system is inconvenient, causing frustration for senior citizens, persons with disabilities, and those short on time due to work or other personal obligations.

A number of residents are unable to access some essential barangay services because of poor digital infrastructure and internet connection within their respective localities [3]. Urban areas might benefit from high-tech internet services, while poor rural barangays are thus unable to participate in digital programs [4]. While some municipalities already share online portals, this cannot be assumed to guarantee that most residents, especially those in rural areas, own smartphones, computers, or stable internet connections [5]. Hence, these services will actually exclude them from the government's e-Governance scheme. Instead of being inclusive, digital

initiatives may further widen the gap between digitally connected and disconnected communities [6].

The effective use of low-cost and power-efficient embedded systems, such as the Raspberry Pi, has enabled the development of low-cost, simple, and power-efficient kiosks [7]. These systems will provide promising opportunities to create digital platforms that work even in areas with limited power and network connectivity. The Smart Barangay Kiosk System will allow residents to request documents and barangay officials to manage resident records without having to use an individual device or an internet connection. The kiosk acts as a digital gateway for the community—easy to use, always available, and not dependent on the resident's access to technology.

In addition to being accessible, this will also simplify administrative processes, lessen the load on staff, and encourage transparency in the community . Not only will staff spend less time manually managing the walk-in requests but they could also carry out actions such as verifying, approving, and updating records by using the system. It also will keep all transactions digitally recorded and reduces the chances of misplacing paper documents or having inconsistent data.

Thus, the present study will focus on the development of an integrated Smart Barangay Kiosk System for document request handling and resident management a step forward in the practical direction of modernizing barangay operations in consonance with national goals of digital transformation. The use of Raspberry Pi in this context emphasizes low-cost innovation that suits the needs and capabilities of local communities [7].

Ever since, municipalities have been enhancing web platforms through which residents may request documents. On the other hand, for those living on the wrong side of the digital divide especially in rural setups or barangays of low-income portals and towns is a big challenge, thereby posing an obstacle for citizens to access essential public services [3]. Digital transformation must consider the availability of both infrastructure and user access. In many barangays, unfortunately, the residents are limited by not only weak connectivity but also the issue of personal digital devices and computer literacy [5]. In the end, many residents are unaware or unable to use online platforms, despite its availability in their area.

It becomes clear that there will be an immediate need for a localized, offline, and very simple-to-use platform through which every resident can access barangay services, taking into account their level of technology access. The need is not just for innovation, but for inclusive innovation—one that reaches every member of the community, regardless of their digital limitations [6].

Under the circumstances, the suggested Smart Barangay Kiosk System will digitize critical barangay functions via a touchscreen kiosk operated on the Raspberry Pi. It enables residents to request documents from a self-reliant system without using their own devices. The kiosk will be installed in public and easily accessible locations such as barangay halls, allowing even the least tech-savvy individuals to navigate basic services.

The kiosk will also have a remote management administrative dashboard that enables authorized barangay officials in the management of resident records and processing of document requests. This dashboard will allow officials to have real-time information about managing more effectively citizen requests and also manage timely and accurate records. Improvement in quality

is not limited to the service area alone, but has also been made for accountability and transparency in public administration [8].

The system will operate in an offline-first manner and integrates into rural barangays with the added advantage of operating effectively even under conditions with very limited or no internet connectivity for long periods. When internet access becomes available, the system will automatically synchronize data with the server, keeping everything updated without needing to upload data manually. This two-way mode (offline and online) increases the reliability of the system.

The combination of all these features with the strength of the Raspberry Pi being a cheap and energy-efficient microcomputer, will make this solution cost-effective, scalable, and sustainable in the long run for community use [7]. This system will be adopted and maintained by small barangays without great budgets and technical staff. It will have compact and energy-efficient properties, making it an ideal system for remote areas with poor or unreliable power supply.

The Smart Barangay Kiosk System aims to enhance access to document services and secure resident data management that will go well with offline functionality to promote that kind of digital inclusion within the community. Through this, technology is also given a role to play in local community efforts toward transparency, access, and delivery of services [8]. By transforming how barangays interact with their constituents, this innovation contributes not only to improved service delivery but also to building trust and participation among residents, strengthening the relationship between the government and the people it serves.

1.2. Statement of the Problem

Manual and repeated processes will continue to undermine the efficiency of barangay offices. Therefore, the residents must go to the barangay hall to request documents. Any processing time is slow, so documents are often misfiled. Certainly, this age-old system will continue to restrict access to handling documentation, especially for the elderly or those unable to travel for work during regular business hours.

Moreover, there is no central digital system where resident data is stored and managed. This lack makes it difficult to track which documents have been requested and verify their records. Limited contact between officials and residents: document requests are mostly handled through in-person visits that are not efficient for all.

This will continue causing slow document processing, inefficient management of the residents' data, and being less transparent, as well as making already difficult access to exclusive digital platforms worse among residents who do not have personal devices or reliable internet access. The absence of a central system for request handling, transaction tracking, or timely availability of public information causes an access inconsistency in service delivery while the administrative workload remains heavy.

That is why there is a call for a cost-efficient, less-power dependent, and scalable solution that can digitize core services of barangays and run even in less connectivity. Such a solution should also improve both resident satisfaction and administrative efficiency when it comes to resolving administrative requests. Additionally, challenges related to system deployment, adoption, and maintenance must be addressed to ensure sustainability at the barangay level.

1.3. Significance of the Study

The Smart Barangay Kiosk System will provide a serious benefit to the lives of residents and barangay councilors alike. By utilizing cheap and available digital technologies, the system will simplify some of the more common administrative tasks causing inefficient transactions and perhaps at least attempt to narrow the digital divide within the community[6].

This innovation will not only simplify service delivery but also promotes transparency, accountability, and inclusivity in local governance. The benefits can be categorized as follows:

For Residents:

- Setting up a user-friendly, freely accessible kiosk that would allow residents to request papers any time they wish without needing to use a computer or mobile phone would be very helpful.
- It will greatly decrease the time and effort required by residents to obtain barangay services.
- Will improve inclusiveness and digital literacy, especially among low-income and elderly communities.

For Barangay Officials:

- Reduced manual labor as a result of centralized record administration and automated processing.
- Issuance of records that are more accountable, transparent, and accurate.
- This enables remote data management and monitoring via a secure admin dashboard.

For Local Government Unit (LGU):

- This supports LGUs' digital transformation initiatives through a scalable, replicable, and cost-effective service delivery model.
- It enables centralized transaction logging and increased accessibility of resident information in enhancing governance efficiency.
- It enhances compliance with national policies on e-governance and modernization programs.

For Future Researchers:

- Developmental groundwork for improved community service kiosks and IoT-based governance tools.
- Provides an opportunity for system expansion: biometric authentication, mobile integration, AI-driven analytics, or multi-barangay deployment.
- Encourages further research for offline-first systems, smart communities, and localized e-governance solutions.

1.4. Objectives

The main objective of the study is to develop, design and implement a Smart Barangay Kiosk System that will modernize and digitize the way barangay services are delivered to the community. With the use of cheap and affordable technology, particularly the Raspberry Pi platform, the system will be able to improve the working efficiency of the government while making accessibility easier to all residents, especially those who have limited access to digital resources, and promote transparency within local governance, according to a project description.

It intended to develop a functional prototype that could also appeal to the residents and barangay officials using the system.

The following are the specific objectives of the study:

1. To develop a kiosk interface that allows residents to safely and easily request barangay documents such as Certificates of Indigency, Barangay Clearances, and other essential services with or without the need for a personal computer or smartphone.
2. To construct a secure and centralized database for storing and managing resident profiles, service requests, and other barangay-related records.
3. To create a web-based administrative dashboard accessible remotely by authorized barangay personnel for real-time management, tracking, and processing of requests, including adding, editing, and searching residents.
4. To automate the printing of a receipt for the walk-in and provision of SMS notifications to alert residents of document availability for collection.
5. To develop offline-first functionality that guarantee continuous system operation during internet outages and allows automatic data synchronization once connectivity is restored.
6. To implement basic user authentication and security measures that help ensure data privacy and prevent unauthorized access.
7. To test and evaluate the system's usability, reliability, and efficiency through deployment in one or two pilot barangays.

1.5. Scope and Delimitations.

This study mainly covers the designing, development, and evaluation of the prototype for the Smart Barangay Kiosk System with Raspberry Pi as the core hardware platform. This study was limited to the following:

- Focuses on designing, developing, and evaluating a Smart Barangay Kiosk System using Raspberry Pi
- Covers core modules: Document Request, Resident Management, Admin Dashboard, Offline-First Operation, Auto-Print Receipts, and SMS Notifications.
- Testing and implementation are limited to Barangay Maangas and Barangay Sta. Maria in Presentacion, Camarines Sur.
- Payment is cash only upon pickup.
- The prototype will support requests for common barangay documents such as the following:
    - ☐ **Barangay Clearance** - is a Philippine government document from your local village (barangay) confirming you're a resident with good moral standing, with no pending issues, needed for jobs, business permits, loans, or opening bank accounts.
    - ☐ **Certificate of Indigency - is** an official document issued by a local government unit (LGU) in the Philippines, typically the barangay or the Municipal Social Welfare and Development Office (MSWDO), to certify that a resident is a member of an indigent or less fortunate family.

- ☐ **Certificate of Residency -** is an official document that verifies an individual is a legal and bona fide resident of a particular locality, such as a village (barangay) or a country.
- ☐ **Certificate of Solo Parent -** is an official document issued by a local government unit (LGU) in the Philippines, in coordination with the Department of Social Welfare and Development (DSWD), that certifies an individual's status as a solo parent under the law.
- ☐ **First-Time Jobseeker Certificate -** is a document from your local Barangay Hall in the Philippines, issued under Republic Act 11261 (First Time Jobseekers Assistance Act), that proves you're a Filipino resident, haven't sought employment before, and are seeking your first job.

The following are the delimitations of the study:

- No integration into any national government system nor any third-party platform.
- No mobile app development except for the Progressive Web App (PWA) version.
- Security is just basic, advanced cybersecurity is not being studied.
- Evaluation is limited to 2 pilot barangays, which may not represent the entire barangay.

CHAPTER 2

RELATED LITERATURE REVIEW AND STUDIES

2.1. Theoretical Basis

This section describes theories which form the core of the design of the Smart Barangay Kiosk System; e-governance, offline-first web architectures, and the principles of human-centered design. These explain in a logical and scientific manner why we were able to develop a low-cost yet highly resilient system that responds to specifics of barangay-level issues in the Philippines such as erratic connectivity and low technical literacy among the population.

2.1.1. E-Governance

E-Governance refers to the employment of information and communication technologies (ICT) meant to improve the delivery of public services and transparency and citizen engagement in governmental processes. This, according to the DILG, further enhances the PDP 2023-2028 through accountability and efficiency in local administrations in barangay-level e-Governance. Agents like the Seal of Good Local Governance for Barangays (SGLGB) and the Retooled Community Support Program (RCSP) demand that digital means be explored for lessening the bureaucratic bottleneck in document requests and resident management systems, thus contributing to the national goal of developing truly inclusive communities [9]. The Smart Barangay Kiosk System further apply this through provisions made for offline document requests, complemented with SMS alerts, resulting in reduced queues and paperwork and ensuring traceability with numbers—the very reason for DILG promoting digitized barangay services in low-resource settings.

2.1.2. Offline-First Progressive Web Apps (PWAs)

Offline-first PWAs encourage online access and service-workers can be used for local storage and caching operations (e.g., IndexedDB), so that synchronization can automatically be resumed once the app comes back online. The architecture is essentially one of progressive enhancement, where core functionalities are targeted for offline use, and enhancements depending on network availability are added for online use [10]. In the Philippine context, where connectivity compromises reliability for rural barangays, they are enabling the development of resilient apps such as the Power Kiosk System. With Supabase employing local cache for offline handling of requests, action is synchronized in real-time, thus assuring complete offline work for residents' submissions and approvals of the administration during the operational phase of the site [11].

2.1.3. HCI for Low-Literacy Users

HCI states that interfaces should seek to be intuitive and minimize cognitive load, especially for a broad range of users such as elderly residents or with limited exposure to technology. A recent study pertaining to rural UX design in the Philippines has advocated multilanguage support (English/Filipino) and a large-button cutout touch interface for maximum accessibility [12]. In the context of the current work, the simple nature of the kiosk forms (name, address, purpose) with receipt printing automatically minimize errors, and the responsive design of the PWA shall be undercut on phones without any login hurdles posed to the residents, thereby increasing usability of the system in barangay halls in case of brownouts and long queues.

2.1.4. Database Management & BaaS

Modern DBMS today have the full capability of ensuring data integrity in real-time while synchronizing offline; for example, Supabase (PostgreSQL-Based BaaS) provides this feature through edge functions with local caching. This is for the Philippine capstone projects for the years 2023-2024 since Supabase would be the most appropriate choice for the scalable yet cheap backends of community apps of which these would have authentication and SMS integrations without the burdens of dealing with heavy server management. In this instance, Supabase is acting on the level of resident records and pending requests in auto-sync mode, thereby maintaining consistency between kiosk and admin dashboard while being compliant with the data privacy restrictions laid out in the Data Privacy Act of 2012 [13].

2.1.5. Raspberry Pi for Embedded Kiosk

Raspberry Pi is placed in an inexpensive single-board computer that is entirely meant to prototype interactive kiosks that operate on Linux, USB peripherals (e.g., thermal printers, etc.), and PWAs via the Chromium browser. In 2023, there was a thesis in the Philippines that deals with Pi-based kiosks that will serve as monitoring instruments for environmental factors and was described as portable and using little energy to operate in off-grid areas. The Pi 5 setup of the project (~₱5,500) can provide sturdy kiosks with UPS support and ESC/POS printing for 24/7 barangay use during outages [14].

2.1.6. System Security and SMS Integration

All security provisions with respect to e-Governance systems have been taken care of regarding authentication, encryption, and role-based access to sensitive resident data. The Philippine guidelines from DILG want secure APIs and hashing for barangay archives. Supabase Auth is the instrument that will log all admin sessions for sending notifications via Semaphore.ph at ₱0.55/SMS with a HIPAA-privacy shield [11].

2.2. Related Studies

This section considers several studies, both foreign and local, about barangay management and kiosks as innovations and gaps. While some prior works did digitization of records, very few integrated full offline PWAs, physical kiosks, or SMS in real deployments-these gaps were addressed by this study.

2.2.1. Community Data Management Using IoT Devices (Foreign)

In the year 2019, Adekunle and Oladipo developed an IoT-enabled kiosk for rural communities in Africa for the self-service submission of forms to be kept locally, with auto-sync. The multilingual interface, aware that the target users may have low literacy skills, emphasized maximum robustness in a low-connectivity environment to limit manual bottlenecks [16].This validates offline kiosks for developing economies, but they do not include SMS or integration with Philippine LGUs for alerts on the readiness of documents, which is integrated into Smart Barangay Kiosk.

2.2.2. Design and Implementation of a Local E-Government Portal (Local)

Reyes and Salonga (2021) have established the online portal for barangay clearance services. They created a section for people that serves as a tracking status and notification in real-time. Although this improves access, not all areas have a constant internet connection, rendering it void. Usability evaluation reduced foot traffic; however, the results showed that hybrid offline systems are necessary. This study builds upon it by using PWA offline caching and kiosk hardware [17].

2.2.3. Barangay Management Information System (Local)

According to Hernandez and Cruz (2021), resident records and permits were digitized and files were created to streamline administering tasks and reduce errors through the use of SQL databases [18]. However, it lacked functionality for self-service by residents or offline modes, thus requiring visitations to the office. This system tends to extend that by allowing no-login requests and thermal printing.

2.2.4. WebYu: A Barangay Yuson Web-based Information Management System (Local)

Santos et al. in 2025 deployed WebYu in Barangay Yuson, Nueva Ecija, using Scrum methodology for automated workflows and with components of user experience (UX) evaluated using Technology Acceptance Model (TAM) having a mean score of 4.74 out of 5. It decreased certification to only 32% of its former duration, but it relied on online access. The idea of the dashboard, therefore, is to enhance resilience through offline PWA and SMS [19].

2.2.5 Development of an Offline-Capable Barangay Health Management System (Local)

In 2024 Lim and Torres had created a mobile PWA for barangay health records, with offline synchronization using Firebase, which could reduce 50% of data entry time across two pilot sites, for health data tracking alone, and meant to work in areas where there is low bandwidth, excluding general requests for documents or printing [20]. This credit goes towards PWA-BaaS hybrids for barangays, and kiosk is an add-on to multi-document workflows using Raspberry Pi hardware [20].

2.2.6 Smart Kiosk for Local Government Services Using Raspberry Pi (Foreign)

Chen et al. (2023) have set up kiosks using the Pi technology in townships across China in order to apply for permits and incorporated touch interfaces with thermal printers for almost all of the time during their operation in low power environments would be 95% [21]. The offline mode managed around 80% of transactions in this setting; however, it required culture-related adjustments for other than Asian contexts. This study localizes it with Supabase and SMS for the Philippines to adapt to the protocols of DILG.

2.2.7 E-Barangay Portal with SMS Integration for Resident Services (Local)

Dela Cruz et al. (2023) developed an online portal for clearances with Twilio SMS notifications resulting in 40% better response rates in urban barangays. Yet, this resulted in a 25% failure rate in their offline use because of outages. Thus, this upholds the SMS utility, which the kiosk-combined full offline PWA offers to ensure there is coverage [22].

2.2.8 Enhanced Barangay Information Management System with Residency Certificate Issuance (Local)

Imus and colleagues (2024) designed a web-based system for the issuance of certificates of residency, which was evaluated via USE questionnaire among officials and IT practitioners (mean score: 4.58/5) [23]. It had the requisite infrastructure for Web-based issuance and no offline kiosk hardware or other document support. This study extends the previous effort by incorporating PWA offline synchronization and thermal auto-printing for walk-ins[23].

2.2.9 One Barangay: A Mobile and Web Barangay Management System (Local)

According to Mirasol et al. (2023), they created a hybrid mobile-web system to centralize record keeping and avoid corrupt data caused by manual Excel usage and tested in urban settings: besides improved security, one disappointment was the need for persistent connectivity, excluding offline conditions prevalent in rural settings [24]. The coupon kiosk addresses this by providing an option for Supabase local cache and Raspberry Pi durability for provisional areas.

2.2.10 Barangay Certificate Issuance and Record Management System with Descriptive Analytics (Local)

Navarro et al. (2025) developed a web application using PHP/MySQL for the analytics of blotter and population reports that enable sorting data by gender/age. While this provided a better understanding for the admin, it lacked the development of self-service kiosks or SMS [25].This informs dashboard analytics extended here with offline PWA requests and real-time notifications.

2.2.11 A Risk Assessment of a Barangay Management System Portal (Local)

The studies made by Fernando et al. (2024) have shown that among the web portals geared towards developmental programs, ICT deficiencies and risks from manual operation pointed out were within Philippine barangays [26].Data handling vulnerabilities are mentioned, but hardware solutions are not recommended. At present, the Pi-based kiosk counter soaks up most of this mess with encrypted offline storage and role-based access [26].

2.3. Synthesis and Research Gap

When synthesizing all these sources, it can be seen that there is a trend in Philippine e-Governance: digitization from simple digitization (Hernandez & Cruz, 2021; Reyes & Salonga, 2021), to partial offline solutions (Lim & Torres, 2024; Santos et al., 2025), inspired partly by international kiosks (Adekunle & Oladipo, 2019; Chen et al., 2023). The new entries along the same lines were Imus, et al. (2024), and Navarro et al. (2025), which leaned more on usability and analytics. Nevertheless, there are still gaps: no integrated system provides 100% offline self-service via PWAs and physical kiosks, along with auto-printing through thermal printers or SMS-that may be deployed in true multi-barangay scenarios. Some methodological insights, such as agile UX testing and risk assessments, informed this 6-month timeline from data collection to analysis. The frameworks of DILG on data privacy also guided this study.

This is the continuation of that chain, filling these gaps-deploying a Supabase-powered PWA on Raspberry Pi in two barangays, supporting cash-only pickups and admin management. This could make findings relevant to national e-Gov goals, potentially indicating scalable tools for rural adoption and addressing connectivity challenges through auto-sync.

CHAPTER 3

METHODOLOGY

3.1. Research Design and Systems Model

The study will follow the Design and Development Research (DDR) method [29], in particular, the prototype development model for the innovation of the Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management. DDR will be appropriate for the study as it involves the design, development, and evaluation of a technology-based solution aimed at improving barangay services through digital transformation, with a focus on offline capabilities to address unreliable internet access in rural areas.

The process includes planning, system design, prototyping, testing, and evaluation. The study will also follow an Iterative Development Life Cycle model consisting of the following phases: Requirements Analysis, System Design, Development and Implementation, Testing and Evaluation, and Refinement and Finalization.

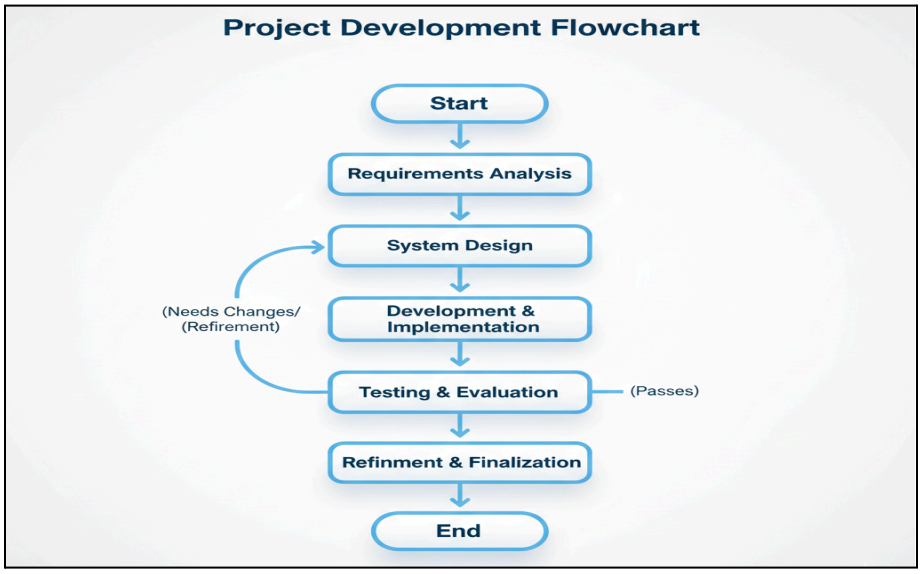

Figure 3.1: Conceptual Framework of the Study

The system model above shows that each phase includes the following: For the Requirement Analysis phase, it involves identifying the project's requirements and objectives. The Data Collection Method gathers data through interviews and surveys, observation, and document analysis that may help the researchers identify the problems needed to address.

The System Design, Development and Implementation phase builds the system using specific development tools and technologies, including hardware components and software. This phase also encompasses the kiosk creation and development process, which involves hardware assembly, software installation and configuration, interface and application development, and thorough kiosk testing.

The system features are based on objectives and are evaluated during the Testing and Evaluation phase. Target respondents and sampling involve identifying participants, such as barangay officials and residents, and determining the sampling technique.

In addition, the model includes Development Boundaries and Prototype Constraints as part of Refinement and Finalization. The model recognizes that there may be restrictions in the development process. Finally, Ethical Considerations are dealt with in order to ensure the accountability and integrity of the entire project.

3.2. Requirement Analysis and Data Collection Methods

In order to guarantee that the Smart Barangay Kiosk System will provide the actual services and address the real needs and challenges of barangay service delivery, multiple data collection techniques will be used. These were selected to be the most relevant qualitative and

quantitative data to inform the design, development, and functionality of the system, particularly that which pertains to an offline-first approach and resident management. The following data collection methods will be employed:

3.2.1. Interviews and Surveys

Structured interviews and guided surveys will be conducted with the selected barangay officials and residents of Barangay Maangas and Barangay Sta. Maria in Presentación, Camarines Sur. The objective of this is to understand the current mechanisms of service delivery from both the administrative and end-user perspectives Barangay officials, such as the secretary, clerks, and the barangay captain, will provide insights into the common challenges they face, including document processing bottlenecks, data storage issues, and communication barriers with residents, as well as needs for offline operation and SMS notifications.

Similarly, residents from diverse backgrounds—including senior citizens, persons with disabilities (PWDs), and low-income individuals—will be surveyed to determine their level of access to digital tools, common service needs (e.g., document requests like Clearance or Indigency), and barriers encountered when availing of barangay services, such as unreliable internet or power outages. The survey will include both closed and open-ended questions to capture both quantitative data such as frequency of service use and waiting times and qualitative feedback such as satisfaction levels, suggestions for improvements, and preferences for features like auto-printed receipts or downloadable receipts.

This method will help identify:

1. Key features the kiosk system must offer, such as offline document requests and resident management.
2. Preferred user interface elements for accessibility, including touch-friendly designs for Progressive Web App (PWA) on kiosks and phones.
3. Gaps in service delivery that can be addressed through automation, like automatic syncing and SMS alerts.

3.2.2 Observation

Direct, non-intrusive observations will be conducted in the Barangay Maangas and Barangay Sta. Maria in Presentacion, Camarines Sur to assess the existing service delivery workflow. The research team will visit barangay halls during regular office hours to observe how documents are requested and processed, how residents interact with staff, how announcements are disseminated, and how records are maintained and stored. This hands-on approach allows the researchers to:

1. Validate the manual and repetitive processes described in the background of the study.
2. Identify physical and workflow limitations that could affect kiosk placement and design, including power reliability for offline operation.
3. Understand peak activity times and patterns of resident engagement.
4. Assess the environment for potential kiosk deployment such as space availability, security, power access, and suitability for hardware like thermal printers and mini UPS.

3.2.3. Document Analysis

To ensure the kiosk system's compatibility with actual administrative Barangay procedures, there will be a thorough review of existing documents and forms, such as:

1. Certificates commonly issued such as Certificate of Indigency and Barangay Clearance.
2. Feedback and complaint forms.
3. Logs of previous document requests and response times.
4. Announcement formats which include printed notices, social media posts.
5. Filing systems such as manual records, Excel sheets, logbooks, etc.

By analyzing these materials, the development team will:

1. Understand required fields and formats for automating documents.
2. Develop accurate digital versions of existing paper forms.
3. Standardize workflows and design database structures based on real data, ,incorporating offline caching.
4. Determine how data privacy is currently handled and what measures need to be added digitally.

The analysis of documents will not prevent the kiosk from enhancing existing work modules to enable smooth and intuitive transition of both staff and residents from a manual to a digital solution, especially with the generation of control numbers and SMS notification features.

3.3. System Design, Development and Implementation Tools and Technologies

The Smart Barangay Kiosk System Development integrates affordable, low-power hardware, and flexible open-source software technologies for cost-effectiveness, scalability, and sustainability. The tools and technologies chosen match the objectives of the project, particularly the need for offline capabilities, easy deployment, and user access in resource-constrained barangay settings.

3.3.1. Hardware Components

1. Raspberry Pi 5 (8GB):

   The Raspberry Pi 5 will be installed in the kiosk as the principal processing unit. It is an inexpensive microcomputer, about the size of a credit card, having enough power and memory to run a local server with full features, managing user interaction, and storing records. Its compact size, low power consumption, and affordability make it ideal for deployment in barangays with limited technical resources or unstable power supply.

2. Touchscreen Display:

   A capacitive touchscreen monitor will be integrated with the Raspberry Pi to provide a user-friendly, intuitive interface for residents. This will allow users to interact with the system through touch-based navigation, removing the need for external input devices like a mouse or keyboard—making it accessible even to users with limited digital experience.

3. Power-efficient Storage (microSD or SSD):

   The system will utilize microSD cards or SSDs to store the operating system, application files, resident data, and transaction records. People often will prefer it

because of the higher endurance and read/write speed of the SSD above the HDD, especially in cases where the applications are frequently used.

4. Other Peripherals
    a. Thermal Printer: A 58mm or 80mm compact, low-maintenance thermal printer will be used to auto-print receipts for walk-in requests, ensuring immediate confirmation with control numbers.
    b. Mini UPS / Power Bank: Will give 2 to 4 hours of power during brownouts for offline functionality running.
    c. Sturdy Case / Stand: For protection, aesthetics, and secure deployment.

3.3.2 Software and Development Tools

1. Frontend Technologies (React + Vite)

   The user interface of the kiosk will be developed as a Progressive Web App (PWA) using React for component-based structure and Vite for fast development and bundling. This enables a single app that works seamlessly on kiosks and phone browsers, with offline support via service workers.

2. Backend and Database (Supabase)

   Supabase (the free tier) would handle backend services, authentication, and database management. It provides offline caching, real-time updating, and automatic syncing upon returning to the internet, thus guaranteeing for fully offline operation of document requests and resident management.

3. Offline Sync

   Supabase local cache combined with service workers will enable automatic data synchronization, preventing data loss during outages.

4. Thermal Printing

   ESC/POS commands via USB (browser or Node.js) for simple and reliable receipt printing.

5. SMS Notification

   Integration with Semaphore.ph or Twilio (₱0.55-₱0.80 per SMS) for cheap, reliable notifications in the Philippines when documents are ready (fee is added in document request pricing).

6. Hosting

   Supabase free tier (500k rows, sufficient for barangay needs) for zero-cost hosting, and up to two active projects (just right for two barangays).

7. Security Measures

   There are basic security features put in place to protect user data while maintaining a certain level of system integrity.

   a. User Authentication: Username + password for barangay staff/admin on the dashboard. Residents require no login for requests.
   b. Access Control: Only authorized personnel can view, edit, or delete records.
   c. Session Management: To protect against unauthorized access, a feature that automatically logs out from a session after a period of inactivity is in place.

3.3.3. Kiosk Creation and System Development Process

The physical and software development of the Smart Barangay Kiosk System will follow a structured, iterative process aligned with the goals of Design and Development

Research (DDR). In this section, we describe the actual activities of work and realization of the kiosk, from hardware assembly and system installation to software development and integration testing.

3.3.3.1 Hardware Assembly

The kiosk will be constructed using cost-efficient components suitable for deployment in low-resource barangay environments. The assembly process will include:

- Mounting the Raspberry Pi 5 within a custom-built kiosk enclosure made of durable and tamper-resistant material.
- Integrating a capacitive touchscreen monitor into the enclosure to serve as the main user interface.
- Connecting peripheral components, including:
    - A thermal printer for auto-printing transaction receipts.
    - A mini UPS for power backup during brownouts.
- Providing internal cable management for USB, HDMI, power supply, and GPIO connections to ensure system stability and safety.
- Power supply configuration using the mini UPS, depending on barangay electricity reliability.
- Ensuring physical accessibility, with the kiosk designed for ease of use by senior citizens and PWDs such as proper screen height, and touch responsiveness.

3.3.3.2 Software Installation and Configuration

The following steps will be performed to prepare the kiosk system for development and use:

- Installing the Raspberry Pi OS on the Raspberry Pi using a microSD card or SSD.
- Setting up Supabase for backend and database, with local caching for offline mode.
- Configuring a local database (SQLite) to store resident profiles, service transactions, and system logs.
- Configuring the PWA with React + Vite to host the kiosk application.
- Installing and configuring software drivers for connected peripherals such as the touchscreen and thermal printer.
- The kiosk application should run automatically at boot by setting autostart scripts, designed so that barangay users can easily use their applications without any technical knowledge.

3.3.3.3 Interface and Application Development

- The user interface (UI) will be developed as a PWA using React + Vite, designed for clarity, simplicity, and accessibility.
- The backend will be handled by Supabase, responsible for:
    - Form submissions (document requests).

    - Local data storage and resident management (add/edit/search).
    - Interaction with peripherals such as thermal printing via ESC/POS.
    - SMS notifications via Semaphore/Twilio.
- Admin Dashboard will be part of the PWA, allowing barangay officials to:
    - Approve or decline document requests (triggering SMS).
    - Add, edit, and search residents.
    - View pending requests in real time.

3.3.3.4 Kiosk Testing and Integration

After development, the kiosk will undergo the following procedures:

- Unit testing of each module (form submissions, printing, data saving, SMS)
- Integration testing to ensure all components (hardware and software) function together seamlessly
- Offline functionality testing to verify that all features work without an internet connection, including data caching and auto-sync.
- Basic security testing, such as checking access control, session timeouts, and login authentication.

3.3.3.5. Offline-First Progressive Web App (PWA) Behavior

The system will be deployed as a Progressive Web App (PWA) using React + Vite + Supabase with service workers. This enables two critical offline scenarios:

1. Kiosk Mode (Raspberry Pi)
    - The kiosk runs the PWA in full-screen kiosk mode on Chromium.
    - All forms, control number generation, resident search, and receipt printing work 100% offline.
    - Requests are cached locally and automatically synchronized when the internet returns.
2. Mobile Phone Access (Resident using own phone – No Internet Needed After First Install)
    - On the first visit (when the resident has even brief internet access), the browser prompts: “Add to Home Screen”.
    - Once installed, the PWA becomes a standalone app on the phone.
    - Residents can open the app anytime (even in airplane mode or no load) to submit document requests.

- Control numbers are generated locally using UUID or timestamp-based logic.
- The requests shall be maintained in IndexedDB (browser local storage) and uploaded automatically, as and when connectivity is established.
- This ensures accessibility even for low-income residents with limited or intermittent data access.

In fact, the offline-first approach was designed to really address the unreliability of internet and electricity in rural barangays, thus making the system truly inclusive and resilient.

3.4. Project Planning and Execution Details

This section gives details on planning and specification of the project which would include the most critical areas that will ensure the successful execution and completion of the project. It provides the features of the system intended for the project objectives, financing regarding project computations and cost-breakdown, including a timeline to direct the research process.

3.4.1. System Features Based on Objectives

The system will consist of the following core modules:

Module	Description
Document Request Module	Allows residents to request documents like Barangay Clearance, Indigency, etc., via walk-in (auto-printed receipt) or online (downloadable/screenshot receipt). Generates instant control numbers.
Resident Management	Enables admin to add, edit, and search registered residents.
Notification System	Send SMS notifications when documents are ready. eg. Your Barangay Clearance is ready for pickup! Control #: BRGY-2025-000123 Payment: P15 Please show this and pay at the barangay hall. Thank you!
Admin Dashboard	Real-time dashboard accessible from phone/laptop for managing requests and residents.
Offline Functionality & Sync	Allows 100% offline operation with automatic syncing when internet returns.

Table 3.4.1: System Features Based on Objectives

3.4.2 Project Cost Estimation and Hardware Breakdown

In order to make the project feasible and sustainable, detailed costing was laid out for the hardware components required for a kiosk unit. The system is intended to be cost-effective, durable, and replicable even by smaller, low-budget barangays.

Estimated cost per functional kiosk (including all the accessories provided for offline operation, receipt printing, and brownout protection) is summarized below:

Item	Approximate Cost (PHP)
Raspberry Pi 5 (8GB RAM)	₱5,500
Capacitive Touchscreen	₱2,000 – ₱3,500
Thermal Printer	₱2,000 – ₱2,500
Mini UPS / Power Bank	₱1,500 – ₱2,000
Sturdy Case / Stand	₱2,000 – ₱3,000
Total Estimated Cost **per** Kiosk	₱13,000 – ₱16,500

Table 3.4.2: Hardware Cost Breakdown per Kiosk Unit

Component	Technology Used	Cost
Frontend & PWA	React + Vite	Free
Backend + Database + Hosting	Supabase (free tier)	Free
Offline Cache & Sync	Supabase + Service Worker	Free
Thermal Printing	ESC/POS via USB	Free
SMS Notifications	Semaphore.ph or Twilio	₱0.55 – ₱0.80 per SMS
Domain (optional)	e.g., kiosk.brgykiosk.ph	~₱500–₱1,000/year (optional)

Table 3.4.3: Software and Operational Cost Summary

Using free-tier cloud services (Supabase), open-source tools, and inexpensive single-board computers assures that this project is very affordable and makes it just repeatable in other barangays in the Philippines without demanding major funding or expertise requirements for maintenance.

3.4.3. System Flowchart

The overall process of the Smart Barangay Kiosk System is illustrated in the flowchart below.

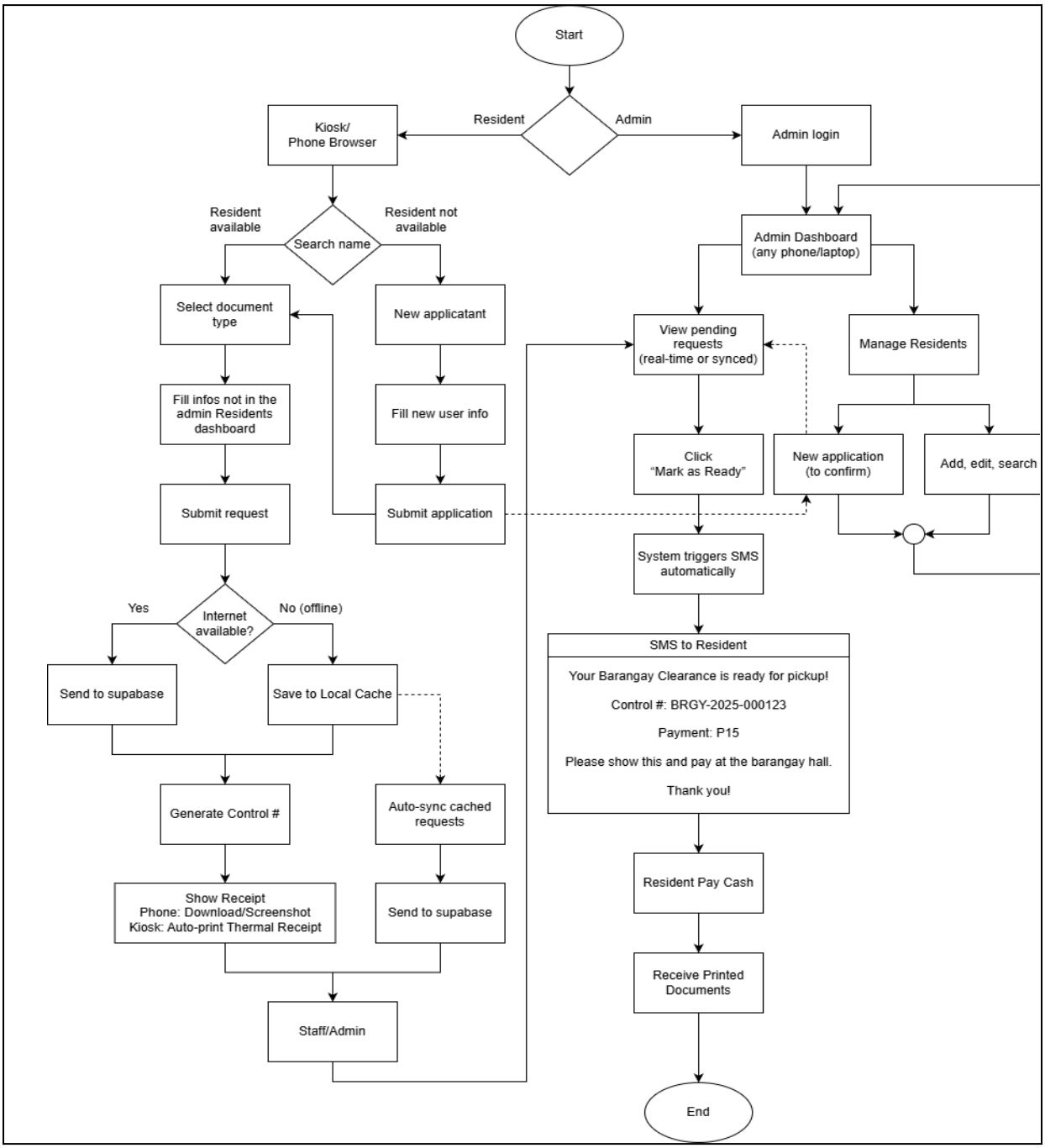

Figure 3.4: System Flowchart of the Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management

3.4.5 Project Timeline and Development Schedule

The entire research, development, and actual deployment of the Smart Barangay Kiosk System in two real barangays will be completed within a realistic six (6)-month period. The schedule follows an iterative and agile approach, allowing flexibility while ensuring all deliverables are met before the final thesis defense.

Month	Activities / Milestones	Deliverables
1	● Conduct interview, survey, observation, and document analysis. ● Finalization of requirements with barangay officials ● UI/UX design and wireframing ● Supabase project setup and domain registration	● Data collection questionnaires and result documentation ● Approved requirements ● Figma prototypes ● Supabase ready
2	● Development of document request form (kiosk & mobile) ● Offline-first functionality & local caching ● Control number generation ● Receipt generation (print + downloadable image)	● Working offline request module
3	● Admin dashboard development ● Resident management (add/edit/search) ● SMS integration (Semaphore/Twilio) ● Thermal printer integration ● Full offline & stress testing (offline/online) ● Hardware assembly and kiosk construction ● Security & backup testing	● Complete admin panel + SMS + printing ● Fully working kiosk prototype
4	● Deployment and staff training in Barangay 1 (Maangas or Sta. Maria) ● Deployment in Barangay 2	● Live system in 1st barangay ● Live system in 2nd barangay
5	● User acceptance testing with real residents ● Bug fixes and minor improvements ● Final Evaluation and Data Gathering	● Evaluation questionnaire and result documentation
6	● Documentation, thesis writing, and presentation preparation ● Thesis defense	● Complete documentation ● Final Thesis Paper

Table 3.4.4: Project Timeline and Major Milestones

The six-month timeline has been validated with actual development progress and barangay coordination. Actual hardware purchases and software development have been scheduled and for some unanticipated delays there had been buffer time incorporated for possibilities of brownouts, connectivity, or scheduling conflicts with the barangay.

3.5. Testing, Evaluation, and Data Analysis

The prototype will be tested in at least two pilot barangays which will be selected based on availability, willingness for participation, and existing infrastructure limitations.

Evaluation methods include:

- Usability Testing: Observing how easily users (residents and staff) interact with the kiosk interface.
- System Performance Testing: Bringing up the reliability of kiosks on both offline and online events, including sync and printing.
- Feedback Collection: Post-use surveys and interviews with users and officials to gather opinions on effectiveness, ease of use, and satisfaction.
- Administrative Efficiency Metrics: Measuring processing time before and after kiosk deployment.

3.6. Target Respondents and Sampling

The system development and evaluation of the Smart Barangay Kiosk System are designed to reflect the real needs of its target users, hence the select group of respondents targeted for the study represents both service providers (barangay officials) and service users

(barangay residents). Such groups are intended for their direct transactions among barangays and for having different access to digital services that is crucial in analyzing the digital divide that kiosks had intended to address.

3.6.1. Barangay Officials and Staff

The first group of respondents includes barangay personnel responsible for daily administrative and public service functions. These may include the Barangay Secretary, Barangay Chairman, Clerks or Administrative Aides and Kagawads involved in public services and technology initiatives.

These officials will provide valuable insights regarding:

- Current service delivery workflows and pain points.
- Commonly requested documents and their processing steps.
- Challenges in managing resident data and paper-based records.
- Preferred features and functionalities in the admin dashboard, such as marking requests ready and SMS triggers.
- Expectations regarding system usability, efficiency, and transparency.

Barangay officials are prime informants on the requirements analysis, system design, and evaluation sessions. Their input will focus much on prioritization of features for the kiosk so that it operates under the realities and constraints of barangay governance.

3.6.2. Barangay Residents

This group comprises residents who are regular users of barangay services such as clearances, certificates of indigency, and other commonly used documents.

These residents will be invited to take part in surveys, usability tests, and focus interviews to determine:

- Ease of navigation through the touchscreen interface.
- Understanding of icons, labels, and instructions.
- Satisfaction with speed and feedback mechanisms.
- What barriers prevented them from using the current online systems.

3.6.3. Sampling Technique

Specific characteristics or other attributes are present in participants purposely selected for inclusion in this study; purposive selection, a nonrandom sampling method, will be applied. The purposive sample assures that the technological functionality and social equity are given consideration in the development of the system and aims to yield a kiosk solution that is technically fit as well as socially responsive and inclusive.

This method is appropriate because:

- The study aims to gather in-depth, targeted insights from individuals who have direct experience with barangay services or are impacted by digital service delivery.
- It ensures the inclusion of underrepresented or marginalized groups who are often overlooked in digital innovation efforts.
- It allows the researchers to select knowledgeable participants who can meaningfully contribute to the design, testing, and evaluation phases of the prototype system.

3.7. Development Boundaries and Prototype Constraints

This study is aimed at design, development, and pilot deployment in at least two real barangays of the Smart Barangay Kiosk System. Essentially, this study aims to create and evaluate a working prototype that can solve core service delivery problems at the barangay level. Therefore, certain technical and operational boundaries are set to maintain project feasibility limits. The following constraints define the limits of the system's development and testing phase:

- Prototype-Stage Deployment: The system will be developed as a functional prototype for pilot testing only. A full-scale rollout or integration into national digital infrastructure is not within the current study's objective.
- Standalone Operation: The kiosk will operate as an independent device. The development will not include integration to national government platforms or even other-third party mobile applications and any future cloud-based system beyond Supabase. No online payment or QR codes will be implemented; payment is cash only upon pickup.
- Security Implementation: The prototype will implement basic security features like user authentication and access control. However, advanced-level security measures such as data encryption, intrusion prevention, and vulnerability assessment will now take the back seat.
- Short-Term Evaluation Focus: Conducting the evaluation shall be limited only to measuring the pilot-testing system's functionality and ease of use and administrative utility. The project is also not intended to assess the long-term impact, sustainability, or cost-effectiveness of large-scale deployment.

These boundaries ensure that the project stays within the technical viability and resource availability parameters, as well as gives an idea of how the digital kiosks will improve service delivery at the barangay levels.

3.8. Ethical Considerations

This research maintains strict compliance with ethical standards for research to guarantee the rights, privacy and welfare of all those involved in designing, developing and evaluating the Smart Barangay Kiosk System. The following procedures will be instituted to ensure ethical integrity for the entirety of this research:

- Informed Consent: An informed consent document, clear and comprehensible to all respondents, will be available before any survey, interview, usability test, or observation session. This document will explain the intent of the study, procedures, possible risks, and potential benefits to the respondents, voluntary participation, and the right to withdraw at any time without penalty. Data collection will include only those individuals who have provided written or verbal consent.
- Confidentiality and Anonymity: Participants' confidentiality will be strictly upheld for any personal detail obtained from those in the study. Responses will be anonymized as necessary, while their identifying details would not be published in any reports or data shared. Further protection of identity may include the usage of pseudonyms or coded identifiers within transcripts and records.
- Data Privacy and Protection: In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173) and applicable data protection regulations, all collected data will be stored in secure, password-protected systems. Only authorized members of the research

team will have access to the raw data. Digital files will be encrypted, and physical documents (if any) will be stored in locked storage. After the completion of the research, data will be retained only for as long as necessary and will be securely disposed of afterward [27].

- Ethical System Design: These will include role-based access control, automatic session logout, and restricted administrative access to make sure that sensitive resident data remains protected. The "smart barangay Kiosk System" will be built on privacy-by-design principles. Furthermore, there is no collection of biometric or highly sensitive personal information unless it is required and consented to.
- Vulnerable Populations: Special consideration will be given to vulnerable groups such as senior citizens, persons with disabilities (PWDs), and low-income residents. The research team will ensure that these individuals are not coerced into participation, and that all research activities are conducted in a respectful, inclusive, and non-discriminatory manner.
- Ethical Review: According to the institutional or academic requirement, if necessary, the research protocol is to be submitted to the Research Ethics Committee (REC) or Institutional Review Board (IRB) for ethical clearance and approval prior to the start of any field activities [28].

By incorporating these principles, it enforces that this study respects, protects and empowers all its participants during the course of research, and advocates responsible innovation and ethical use of new media in public service.

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